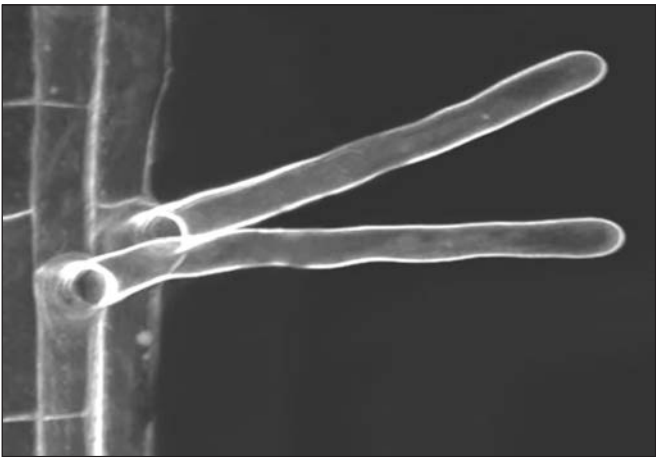


8. The pictures below show a young bean seedling (*Vicia faba*) root and an electron micrograph of root hair cells.



Root with root hairs
(left) and magnified
view of root hairs (right)



(a) State how the root hair cell is adapted for its function. [1]

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(b) Describe how water moves from the soil into a root hair cell. [4]

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Examiner
only

The following is an extract from a UK government report from 2016:

“During the period from mid-December 2013 through until early February 2014 the UK suffered a spell of extreme weather, with a series of very large winter storms battering the country. Much of the UK suffered from extensive flooding during January 2014; this period saw the highest rainfall on record. The floods impacted on the agricultural sector through damage to or loss of established crops.”

Soil becomes waterlogged when water builds up and is unable to drain away. This reduces the oxygen levels available to root hair cells.

- (c) Explain why reduced oxygen levels for the root hair cells results in poor growth and yellowing of the leaves. [6 QER]

END OF PAPER

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